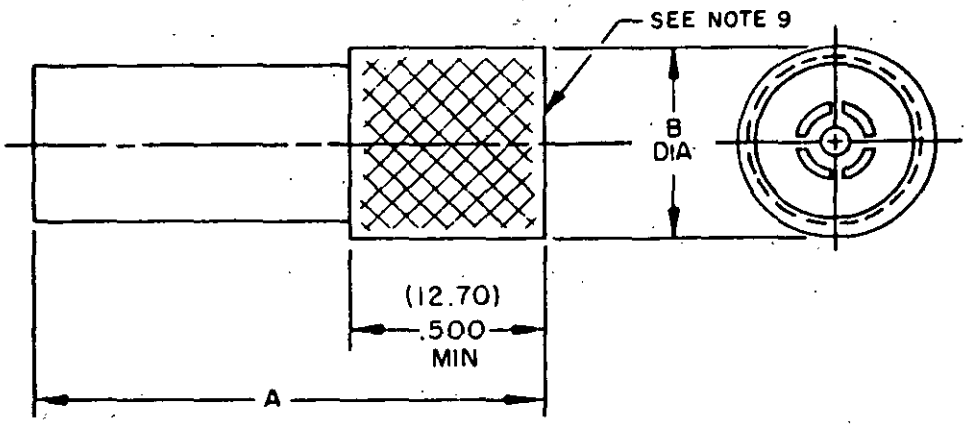
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES N (CABLED), PIN CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE FOR CATEGORY B, SEE TABLE IV



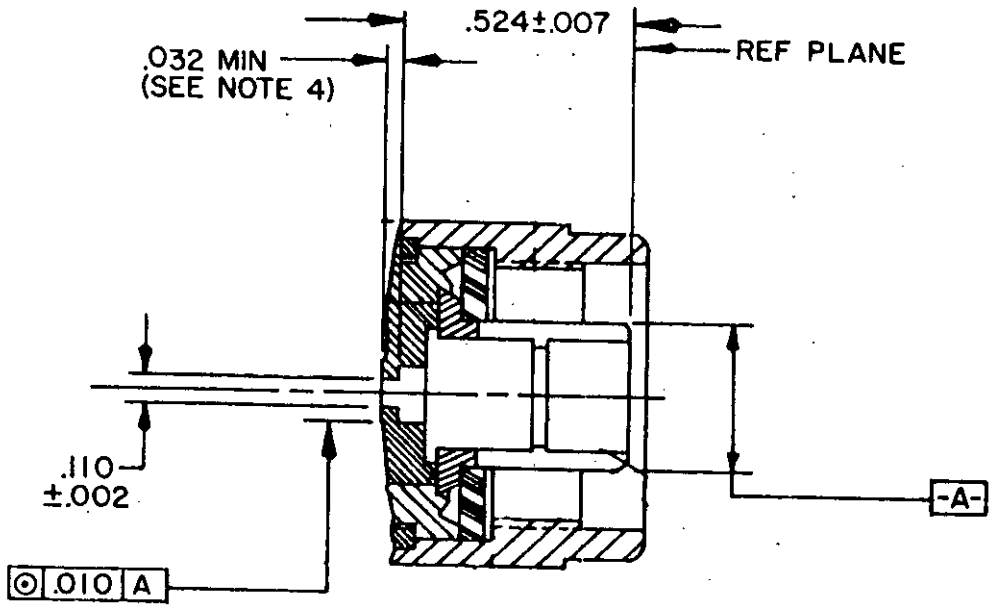
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. See table I and III for dimensions A and B.
5. Dimensions A and B are the largest overall dimensions of the connector.
6. Wrench flats are to accommodate standard wrench opening in accordance with H-28, appendix 10.
7. All undimensioned pictorial representations are for reference purposes only.
8. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
9. Series N, pin contact interface in accordance with MIL-STD-348, 304.1.

FIGURE 1. General configurations.

AMSC N/A
FSC 5935

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CATEGORY D

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.060	1.52	.130	3.30	.230	5.84
.003	0.08	.063	1.60	.158	4.01	.330	8.38
.007	0.18	.066	1.68	.168	4.27	.398	10.11
.010	0.25	.110	2.79	.170	4.32	.412	10.46
.016	0.41	.119	3.02	.177	4.50	.524	13.31
.032	0.81	.124	3.15	.197	5.00	.625	15.88
				.210	5.33	.630	16.00
						.827	21.01

NOTES:
1. Dimensions are in inches.
2. Three holes .027 (0.69 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut optional.
3. Metric equivalents are given for general information only.
4. Chamfer is optional, if chamfer is used put chamfer on a 30° maximum.

FIGURE 2. Mating dimensions for socket terminations.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash no. 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) maximum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0101 3/	2-RG6 4/ 73-RG2T2 5/ 162-00001	A B	1.875(47.63) .827(21.01)
0104 3/	79-RG218 5/	A B	2.500(63.50) 1.343(34.11)
0005 3/	6-RG11 4/ 62-RG144 4/ 65-RG165 74-RG213 75-RG214 5/ 77-RG216 4/ 86-00001 6/	A B	1.875(47.63) .827(21.01)
0015 3/	78-RG217 5/	A B	1.875(47.63) .906
0125 3/	74-RG215 5/ 7/	A B	2.500(63.50) .827(21.01)
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 8/ SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE			
0016 3/	73-RG212 5/ 9/ 162-00001 9/	A B	2.000(63.50) .827(21.01)
0017 3/	74-RG213 5/ 10/		
0018 3/	75-RG214 5/ 10/		
0021 3/	65-RG165 5/ 6/ 10/	A B	2.000(63.50) .827(21.01)
0022 3/	86-00001 5/ 6/ 10/		
0023 3/	6-RG11 4/ 10/ 62-RG144 4/ 5/ 6/ 10/		
0024 3/	77-RG216 4/ 5/ 6/ 10/		
0027 3/	2-RG6 4/ 5/ 9/		

See footnotes at end of table.

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TABLE I. Dash numbers, cross-reference, and dimensions - Continued.

Dash no. 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) maximum
CATEGORY D - FIELD REPLACEABLE - DEFINED PIECE PARTS 8/11/12/			
0501 3/	75-RG214 5/ 127-RG393 86-00001	A B	1.593(40.48) .827(21.01)
0502 3/	74-RG213 5/		
0503 3/	60-RG142 5/ 6/ 128-RG400	A B	1.516(38.50) .827(21.01)
0504 3/	28-RG058		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

4/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

5/ Cable to be used when performing tests requiring cable except as in 4/ and 6/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ Armored cable.

8/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

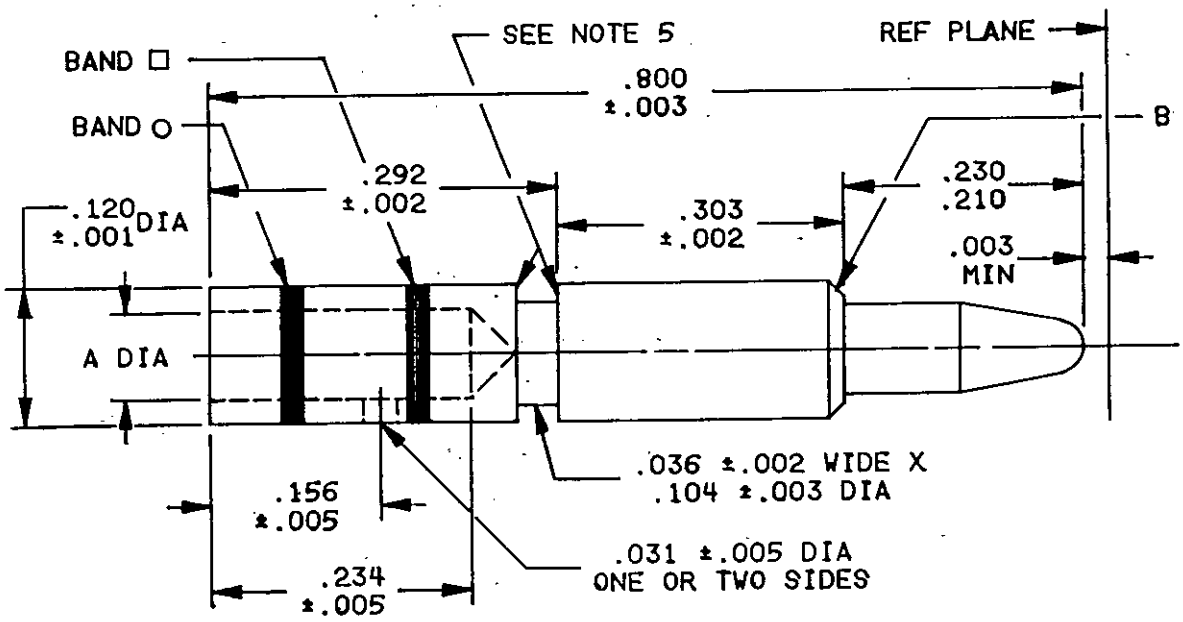
9/ M22520/5-35 Closure A or M22520/5-55 Closure A

10/ M22520/5-61

11/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

12/ Not to be used in Army equipment.

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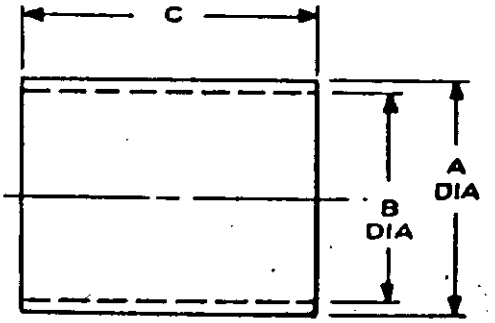
CENTER CONTACT

Inches	mm	Inches	mm
.001	0.03	.210	5.33
.002	0.05	.220	5.59
.003	0.08	.230	5.84
.005	0.13	.234	5.94
.006	0.15	.250	6.35
.015	0.38	.292	7.42
.031	0.79	.303	7.70
.036	0.91	.418	10.62
.043	1.09	.438	11.13
.098	2.49	.492	12.50
.104	2.64	.500	12.70
.120	3.05	.600	15.24
.156	3.96	.800	20.32

Dash no.	Contact no. 2/	A	B	Basic crimp tool 1/ M22520/1	Crimp die or positioner M22520/1	Crimp tensile min	Color band □	Color band ○
0501 0502	1-10	.098 ±.002	.006 ±.002 X 45° ±5°	-01	-13	60 lbs	Red	Black
0503 0504	1-11	.043 +.001 -.002	.006 ±.002 X 30° ±5°	-01	-13	20 lbs	Blue	

FIGURE 3. Contact and ferrule dimensions for category D only.

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CRIMP FERRULE

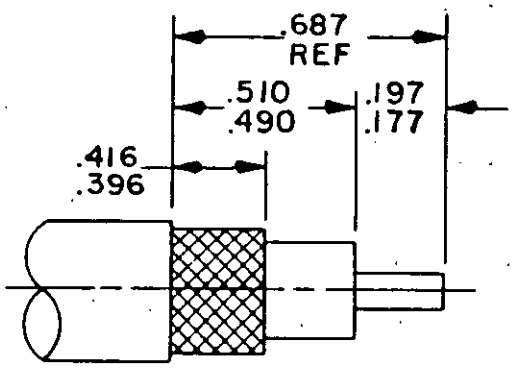
Dash no.	Ferrule no. 2/	A	B	C	Basic crimp tool 1/ M22520/5	Crimp die or positioner M22520/5-
0501	1-50	.492 +.003 -.000	.438 +.003 -.000	.600 ±.015	-01	61 or 25 Closure A
0502	1-51		.418 +.003 -.000			
0503	1-52	.250 ±.003	.220 ±.003	.500 ±.015		05, 11, 57 Closure A or 19 Closure B
0504	1-53	.245 ±.003	.206 ±.003			

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. .003 maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,000 volts rms, maximum working voltage at sea level.

250 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 6 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds minimum.

Inspection conditions: Coupling torque - 6 to 10 inch-pounds.

Mating characteristics:

See MIL-STD-348 and figure 2 for dimensions.

Outer contact:

Test ring ID - .316 maximum, 16 microinch finish.

Insertion force - 25 pounds maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only). 15 pounds minimum axial force for all cable except RG400 and RG142; 6 lbs minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.008 + .002 F (F in GHz).

Item 16 - VSWR shall be less than 1.008 + .002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.012 + .004 F (F in GHz).

Group B inspection - VSWR shall be less than 1.024 + .007 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.08.

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Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum:

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.2	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,500 volts rms minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock : Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 500 volts, minimum.

Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds minimum.

Crimp assemblies:

50 pounds minimum for cables .155-.189 OD.

60 pounds minimum for cables .190-.229 OD.

75 pounds minimum for cables .230-.249 OD.

90 pounds minimum for cables .250 OD and larger.

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .15 dB maximum tested at 10 GHz.

Part number: M39012/01- (dash number from table I or "B" number from table III).

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TABLE II. Group Qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M39012/01 -0101 -0104 -0125	M39012/01 -0101 -0104 -0125
II	-0005 -0015	-0005 -0015
III	B0006 B0007 B0008 B0011 B0012	B0006 B0007 B0008 B0011 B0012
IV	-0016 -0017 -0018 -0021 -0022	B0006 B0007 B0008 B0011 B0012 B0013 B0014 -0016 -0017 -0018 -0021 -0022 -0023 -0024
V	B0013 B0014	B0013 B0014
VI	-0023 -0024 -0027	B0013 B0014 -0023 -0024 -0027
VII	B0026	B0026
VIII	-0501 -0502	-0501 -0502 -0503
IX	-0503	-0503
X	-0504	-0504

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY.

Part number M39012/01B 1/ 2/	Applicable cable 3/ M17/	Dimensions	Inches (millimeters) maximum
0006 4/	73-RG212 5/ 162-00001	A B	2.000(50.80) .827(21.01)
0007 4/	74-RG213 5/		
0008 4/	75-RG214 5/		
0011 4/	65-RG165 5/ 6/	A B	2.000(50.80) .827(21.01)
0012 4/	86-00001		
0013 4/	6-RG11 7/ 62-RG144 5/ 6/ 7/		
0014 4/	77-RG216 5/ 7/		
0026 8/	2-RG6 6/ 7/		

1/ For cross-reference of dash number to superseded part number or type designation, see table IV.

2/ For maintenance replacements for category B, see table V.

3/ The latest version of each cable shall be applicable.

4/ Inactive for new design.

5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

8/ These connectors have captivated center contacts.

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TABLE IV. Cross reference of part numbers. 1/

Preferred part number M39012/01	Substitute for part number or type designation 2/3/
-0101	MS91231-18E, UG-18/U, M39012/01-0001
-0005	MS91236-21G, UG-21/U, M39012/01-0002
-0015	MS91238-204D, UG-204/U, M39012/01-0003
-0104	MS91239-167F, UG-167/U, M39012/01-0004
-0005	MS90534-1185A, UG-1185A/U
B0006	M39012/01-0010, UG-1680, UG-1684, M23329/2-01, M39012/01-0006 M23329/2-06
B0007	M23329/2-02, UG-1681, M39012/01-0007
B0008	M23329/2-03, UG-1682, M39012/01-0008
B0011	M23329/2-07, UG-1685, M39012/01-0011
B0012	M23329/2-08, UG-1695, M39012/01-0012
B0013	M23329/2-09, UG-1817, M39012/01-0013
B0014	UG-1817/U, M39012/01-0014
-0015	UG-1487A/U
-0016	M39012/01-0020
-0017	
-0021	
-0022	
-0023	
-0024	
-0125	MS90293-941C, UG-941, M39012/01-0025
B0026	M39012/01-0026
-0027	
-0501	
-0502	
-0503	
-0504	UG-536/U

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged.

2/ The superseded part number or the type designation is for cross-reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/01-XXXX shall be used in all cases for marking and identifying the connector.

3/ The basic type designation includes all letter versions of the specificed number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

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TABLE V. Maintenance replacements for category B.

Category B no. 1/	Category C dash no.	Category A dash no.	Category D dash no.
B0006	0016	0101	---
B0007	0017	0005	0502
B0008	0018	0005	0501
B0011	0021	0005	---
B0012	0022	0005	0501
B0013	0023	0005	---
B0014	0024	0005	---
B0026	0027	0101	---

1/ Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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